Alternative Assessment Tool (AAT) Report on

“Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL”

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

[Course Code: 23ECSPE1AD]

Submitted by

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Submitted to

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BMS College of Engineering

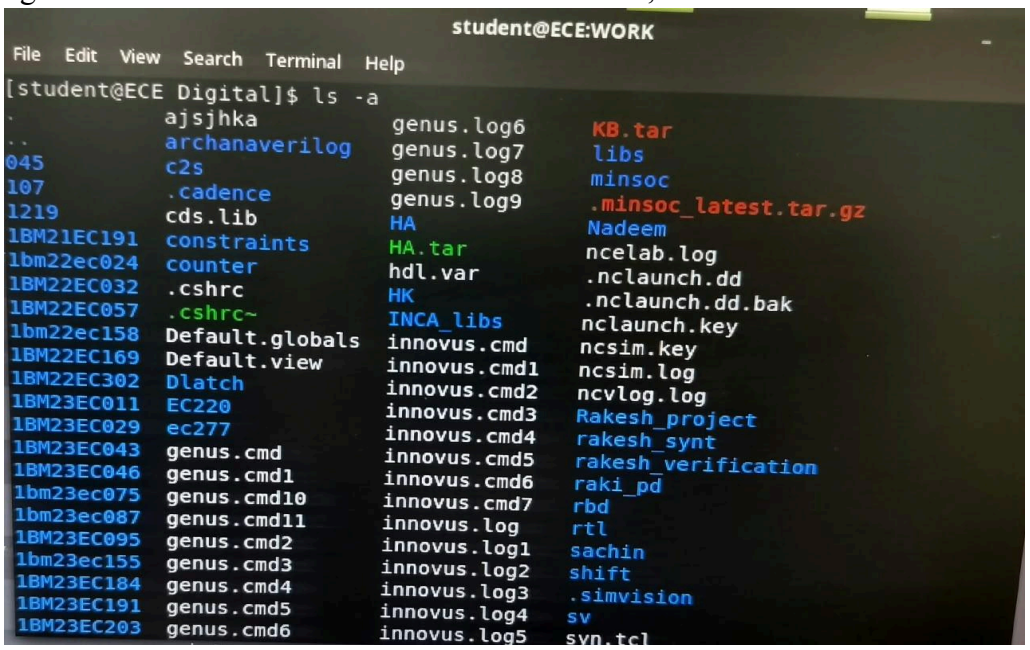
Bull Temple Road, Basavanagudi, Bengaluru-560019

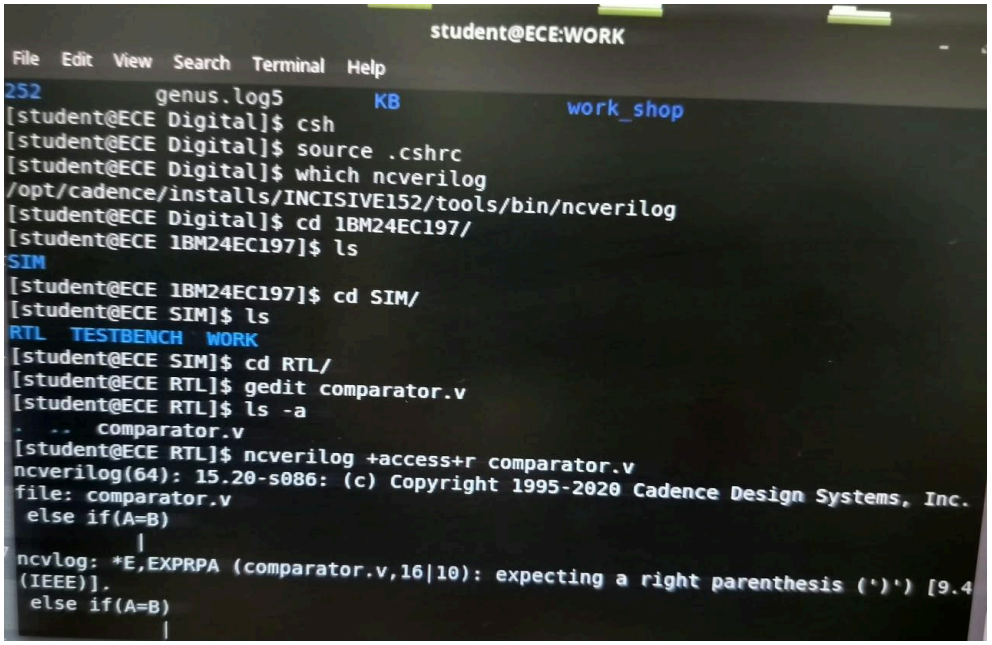
Academic Year

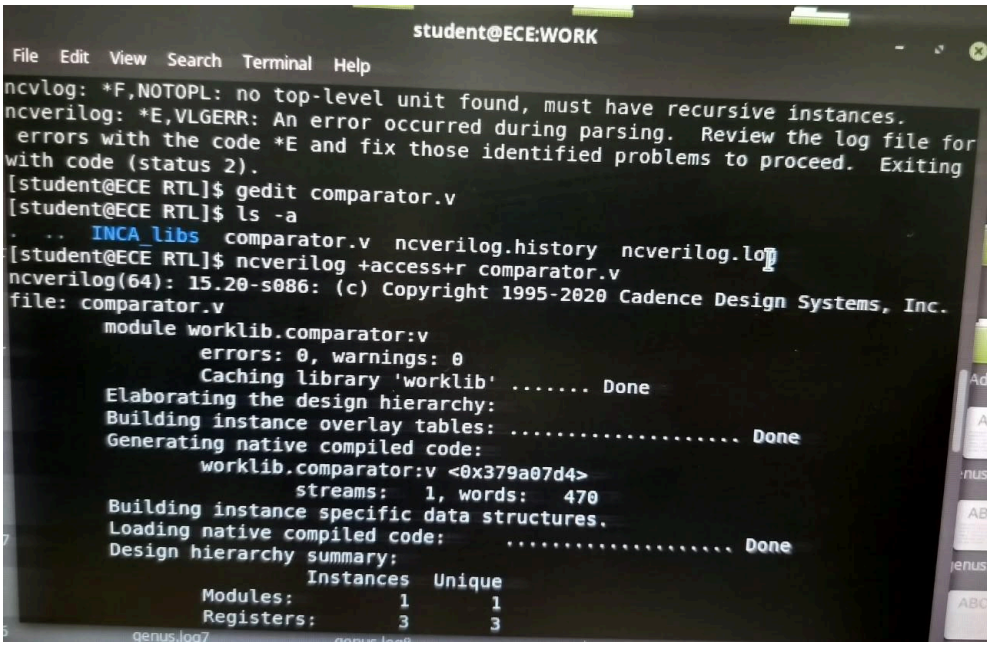
2026 – 2027

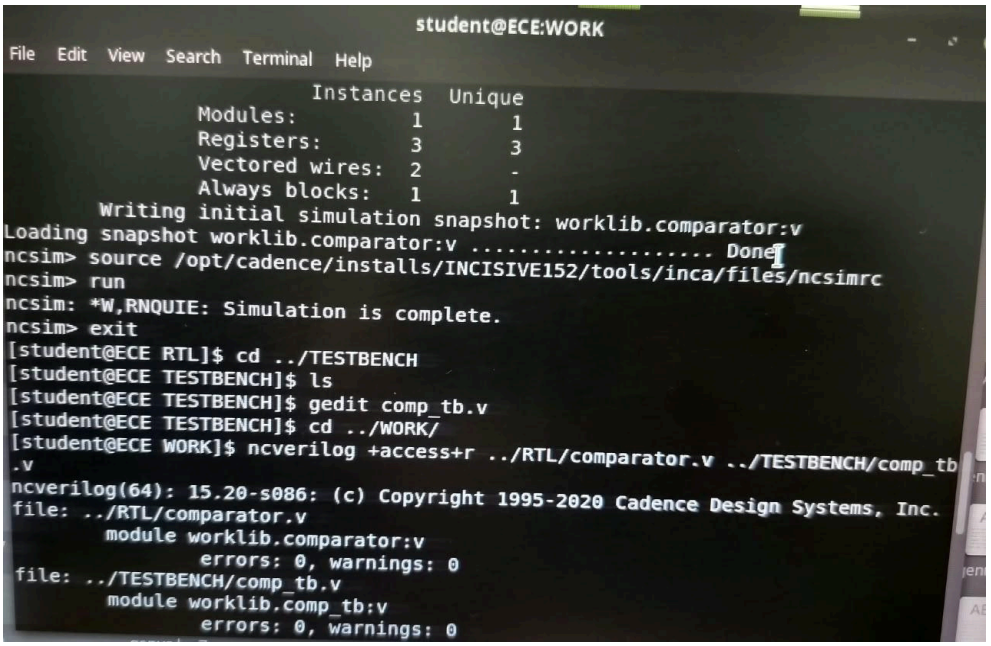
CADENCE SIMULATION STEPS:

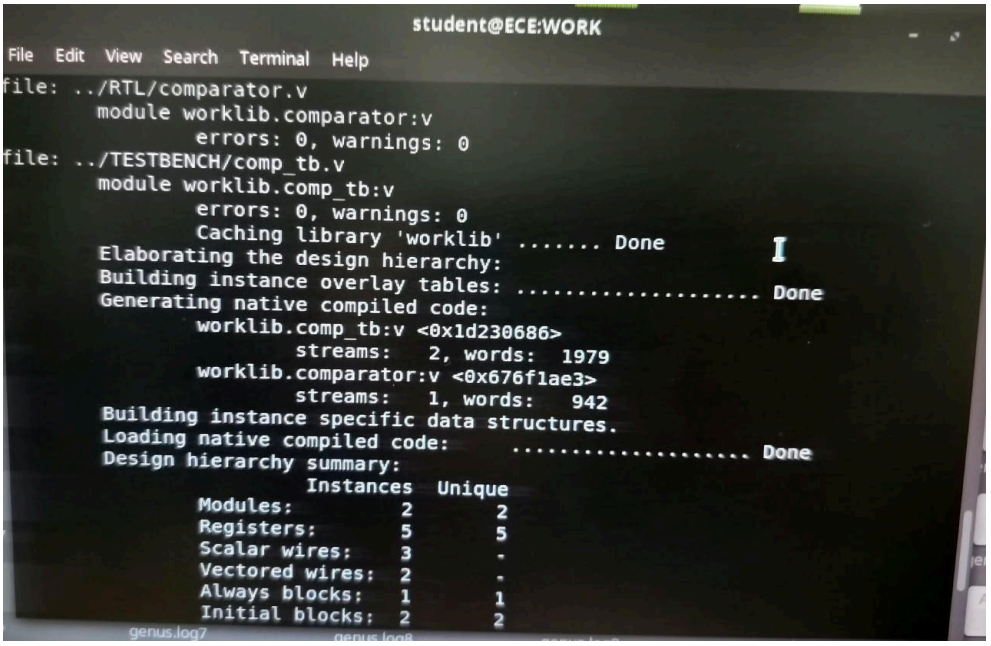
- Open the terminal, switch to the C shell and set up the Cadence environment:
  csh
  source .cshrc
- Verify that the NCVerilog tool is available:
  which ncverilog
- Go to the project directory, which contains the folder RTL,TESTBENCH and WORK:
  cd 1BM24EC197/SIM
  ls
- Go to the RTL folder and write the Verilog design code:
  cd RTL/
  gedit <module>.v
- Go to the TESTBENCH folder and write the testbench code:
  cd ../TESTBENCH
  gedit <tb>.v
  testbench needs `$dumpfile("<module>.vcd");` and `$dumpvars(0, <tb>);` inside an initial block so that the `.vcd` file is created with the name
- Go to the WORK folder, where all simulation outputs are stored:
  cd ../WORK
- Compile, elaborate and simulate the design along with its testbench:
  ncverilog +access+r ../RTL/<module>.v ../TESTBENCH/<tb>.v
- The log shows errors : 0 ; warnings : 0; for the design and the testbench, the design hierarchy is elaborated.
- List the files in the WORK folder and confirm that the waveform file `<module>.vcd` has been generated :
  ls
- Open the waveform in SimVision:
  simvision <module>.vcd
- In SimVision, select the testbench signals (inputs, clock and reset if present, and outputs), right-click and choose Send to Waveform Window, then click Zoom Fit

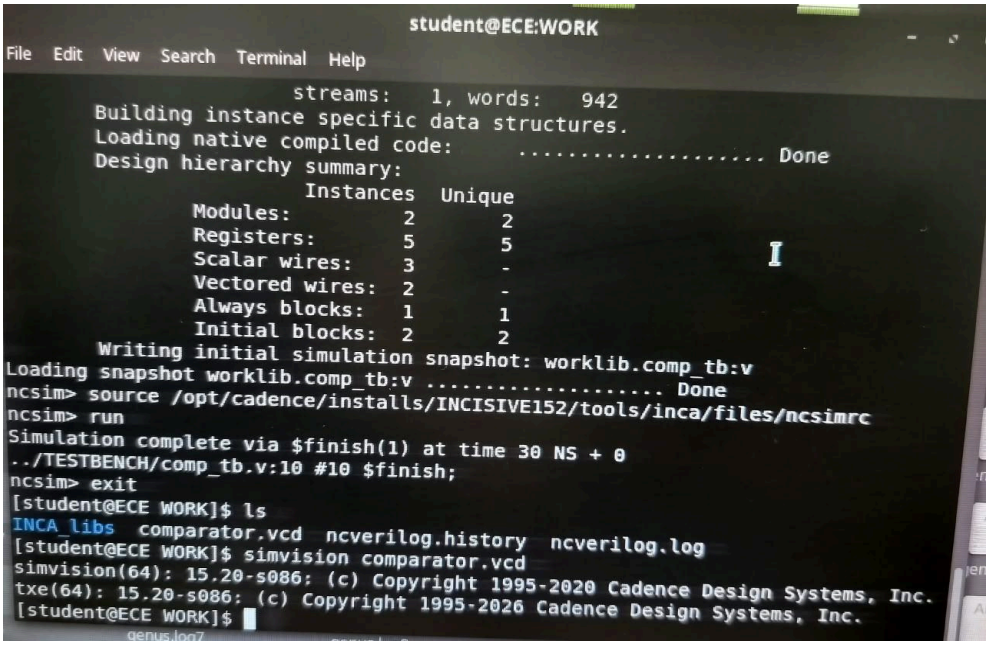



student@ECE:WORK
File Edit View Search Terminal Help
252 genus.log5 KB work_shop
[student@ECE Digital]$ csh
[student@ECE Digital]$ source .cshrc
[student@ECE Digital]$ which ncverilog
/opt/cadence/installs/INCISIVE152/tools/bin/ncverilog
[student@ECE Digital]$ cd 1BM24EC197/
[student@ECE 1BM24EC197]$ ls
SIM
[student@ECE 1BM24EC197]$ cd SIM/
[student@ECE SIM]$ ls
RTL TESTBENCH WORK
[student@ECE SIM]$ cd RTL/
[student@ECE RTL]$ gedit comparator.v
[student@ECE RTL]$ ls -a
. .. comparator.v
[student@ECE RTL]$ ncverilog +access+r comparator.v
ncverilog(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
file: comparator.v
else if(A=B)
|
ncvlog: *E,EXPRPA (comparator.v,16|10): expecting a right parenthesis (')') [9.4
(IEEE)].
else if(A=B)
|


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File Edit View Search Terminal Help
ncvlog: *F,NOTOPL: no top-level unit found, must have recursive instances.
ncverilog: *E,VLGERR: An error occurred during parsing. Review the log file for
errors with the code *E and fix those identified problems to proceed. Exiting
with code (status 2).
[student@ECE RTL]$ gedit comparator.v
[student@ECE RTL]$ ls -a
. .. INCA_libs comparator.v ncverilog.history ncverilog.log
[student@ECE RTL]$ ncverilog +access+r comparator.v
ncverilog(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
file: comparator.v
module worklib.comparator:v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Building instance overlay tables: ..... Done
Generating native compiled code:
worklib.comparator:v <0x379a07d4>
streams: 1, words: 470
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 1 1
Registers: 3 3

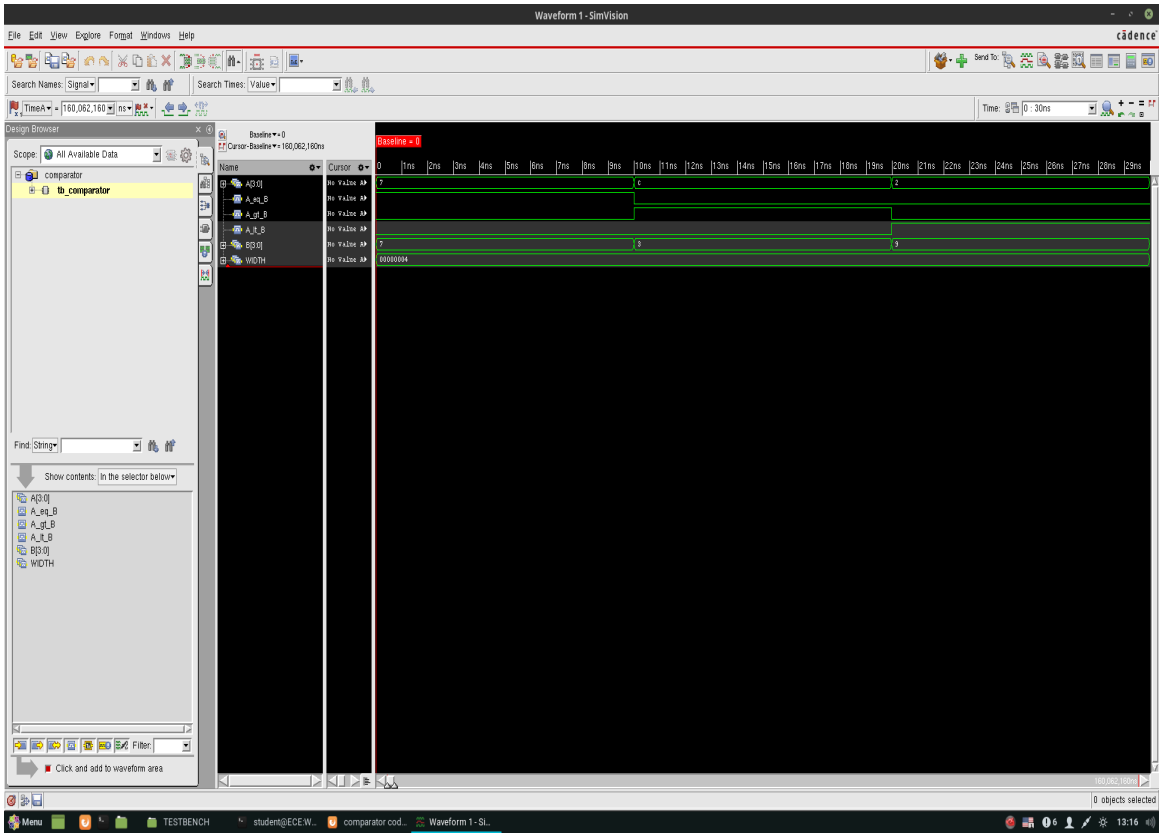

student@ECE:WORK
File Edit View Search Terminal Help
Instances Unique
Modules: 1 1
Registers: 3 3
Vectored wires: 2 -
Always blocks: 1 1
Writing initial simulation snapshot: worklib.comparator:v
Loading snapshot worklib.comparator:v ..... Done
ncsim> source /opt/cadence/installs/INCISIVE152/tools/inca/files/ncsimrc
ncsim> run
ncsim: *W,RNQUIE: Simulation is complete.
ncsim> exit
[student@ECE RTL]$ cd ../TESTBENCH
[student@ECE TESTBENCH]$ ls
[student@ECE TESTBENCH]$ gedit comp_tb.v
[student@ECE TESTBENCH]$ cd ../WORK/
[student@ECE WORK]$ ncverilog +access+r ../RTL/comparator.v ../TESTBENCH/comp_tb
.v
ncverilog(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
file: ../RTL/comparator.v
module worklib.comparator:v
errors: 0, warnings: 0
file: ../TESTBENCH/comp_tb.v
module worklib.comp_tb:v
errors: 0, warnings: 0


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File Edit View Search Terminal Help
file: ../RTL/comparator.v
module worklib.comparator:v
errors: 0, warnings: 0
file: ../TESTBENCH/comp_tb.v
module worklib.comp_tb:v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Building instance overlay tables: ..... Done
Generating native compiled code:
worklib.comp_tb:v <0x1d230686>
streams: 2, words: 1979
worklib.comparator:v <0x676f1ae3>
streams: 1, words: 942
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 5 5
Scalar wires: 3 -
Vectored wires: 2 -
Always blocks: 1 1
Initial blocks: 2 2


student@ECE:WORK
File Edit View Search Terminal Help
streams: 1, words: 942
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 5 5
Scalar wires: 3 -
Vectored wires: 2 -
Always blocks: 1 1
Initial blocks: 2 2
Writing initial simulation snapshot: worklib.comp_tb:v
Loading snapshot worklib.comp_tb:v ..... Done
ncsim> source /opt/cadence/installs/INCISIVE152/tools/inca/files/ncsimrc
ncsim> run
Simulation complete via $finish(1) at time 30 NS + 0
../TESTBENCH/comp_tb.v:10 #10 $finish;
ncsim> exit
[student@ECE WORK]$ ls
INCA_libs comparator.vcd ncverilog.history ncverilog.log
[student@ECE WORK]$ simvision comparator.vcd
simvision(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
txe(64): 15.20-s086: (c) Copyright 1995-2026 Cadence Design Systems, Inc.
[student@ECE WORK]$

1) Comparator

SIMULATION OUTPUT:



2)Johnson counter

SIMULATION OUTPUT:

